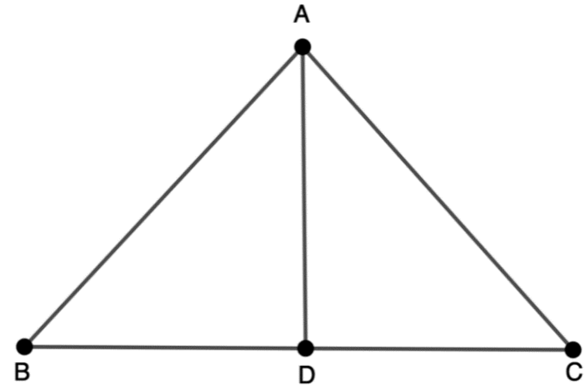
Geometry
Unit 2
Lesson 8 Practice

Name____________________
Date____________________
Period__________________

Complete the two-column proof.

Given: $\angle ADB$ is a right angle, $\angle ADB$ and $\angle ADC$ are supplementary.

Prove: $\angle ADB \cong \angle ADC$

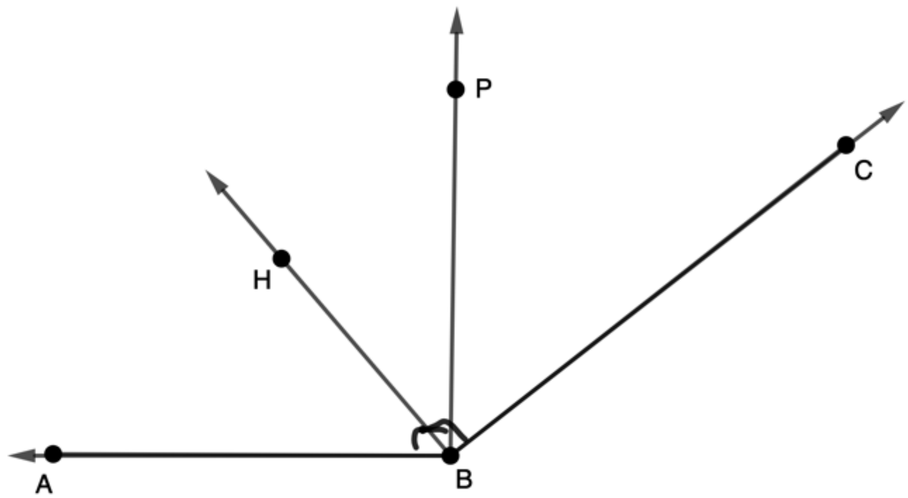


	Statement	Reason
1.	$\angle ADB$ is a right angle	Given
2.	$m\angle ADB = 90°$	Definition of right angle
3.	$\angle ADB$ and $\angle ADC$ are supplementary	Given
4.	$m\angle ADC + m\angle ADB = 180°$	Definition of supplementary angles
5.	$m\angle ADC + 90° = 180°$	Substitution Property
6.	$m\angle ADC = 90°$	Subtraction Property
7.	$m\angle ADC = m\angle ADB$	Transitive Property of Equality
8.	$\angle ADC \cong \angle ADB$	Definition of Congruence

Complete the paragraph proof to answer questions 5-10.

Given: $\angle PBA$ and $\angle HBC$ are right angles

Prove: $\angle ABH \cong \angle PBC$



It is given that $\angle PBA$ and $\angle HBC$ are right angles. Using the addition postulate,

6. $m\angle ABH + m\angle HBP = m\angle PBA$ and 7. $m\angle HBP + m\angle PBC = m\angle HBC$.

Using the 8. transitive property of equality, we know

$m\angle ABH + m\angle HBP = m\angle HBP + m\angle PBC$. Next, we apply the subtraction property of equality

to determine 9. $m\angle ABH = m\angle PBC$. By the definition of 10. congruence,

$\angle ABH \cong \angle PBC$.